# Jongseo Lee

Researcher, Vision and Learning Lab, Kyung Hee University, South Korea

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## RESEARCH INTERESTS

- **Trustworthy video understanding:** interpretable concept-based models; diagnosing and improving Video-LLMs (motion and physical reasoning)
- **Video world models for Physical AI:** learning to perceive and predict physical dynamics from video (intuitive physics)
- **Video representation learning:** multimodal (audio-visual, video-language) and continual learning

## EDUCATION

**M.S. in Computer Science and Engineering**, Kyung Hee University — Sep. 2023 – Aug. 2025
Advisor: Prof. Jinwoo Choi · GPA 4.21 / 4.3

**B.S. in Biomedical Engineering & Electronic Engineering**, Kyung Hee University — Mar. 2017 – Aug. 2023
Double major · GPA 3.95 / 4.3

## PUBLICATIONS

** equal contribution † corresponding author*

*Conferences & Journals*

1. **Jongseo Lee***, Hyuntak Lee*, Sunghun Kim, Sooa Kim, Jihoon Chung, Jinwoo Choi†, Which Way Did It Move? Diagnosing and Overcoming Directional Motion Blindness in Video-LLMs. Accepted to *Neural Information Processing Systems* (**NeurIPS**) 2026. Also at the CompLearn Workshop, ICML 2026 (non-archival). paper project code
2. Wooil Lee*, Junpyo Hong*, **Jongseo Lee**, Tuomas Oikarinen, Tsui-Wei Weng, Jinwoo Choi†, TRACE: Trajectory-Aware Concepts for Explainable Video Understanding. Accepted to *Neural Information Processing Systems* (**NeurIPS**) 2026.
3. **Jongseo Lee**, Wooil Lee, Gyeong-Moon Park†, Seong Tae Kim†, Jinwoo Choi†, Disentangled Concepts Speak Louder Than Words: Explainable Video Action Recognition. *Neural Information Processing Systems* (**NeurIPS**) 2025, *Spotlight* (top 3.5%). paper code
4. **Jongseo Lee***, Kyungho Bae*, Kyle Min, Gyeong-Moon Park†, Jinwoo Choi†, ESSENTIAL: Episodic and Semantic Memory Integration for Video Class-Incremental Learning. *IEEE/CVF International Conference on Computer Vision* (**ICCV**) 2025, *Highlight* (top 2.9%). paper code
5. **Jongseo Lee***, Joohyun Chang*, Dongho Lee, Jinwoo Choi†, CA$^2$ST: Cross-Attention in Audio, Space, and Time for Holistic Video Recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (**TPAMI**) 2026. paper
6. **Jongseo Lee**, Wooil Lee, Gyeong-Moon Park†, Seong Tae Kim†, Jinwoo Choi†, PCBEAR: Pose Concept Bottleneck for Explainable Action Recognition. *CVPR Workshop on Explainable AI for Computer Vision* (**XAI4CV**) 2025, *Spotlight* (top 16.7%). paper
7. **Jongseo Lee***, Dongho Lee*, Jinwoo Choi†, CAST: Cross-Attention in Space and Time for Video Action Recognition. *Neural Information Processing Systems* (**NeurIPS**) 2023. paper poster code

*Preprints*

8. **Jongseo Lee***, Geo Ahn*, Seong Tae Kim†, Jinwoo Choi†, PCEvE: Part Contribution Evaluation Based Model Explanation for Human Figure Drawing Assessment and Beyond. *arXiv preprint*, 2024. paper

*Non-Archival Workshop Presentations*

– **Jongseo Lee***, Joohyun Chang*, Dongho Lee, Jinwoo Choi†, BiAdA: Holistic Video Recognition via Bidirectional Adapter Fusion. *ICCV Workshop on Long Video Foundations* (**LongVidFoundations**) 2025, poster.

*Domestic Publications*

– **Jongseo Lee**, Su Hyeon Kim, Sun Woong Jang, Jun Yeong Moon, Doug Young Suh†, Metaverse Interface with Haptic and Rigid Sense Feedback at a Low Cost. *Journal of Appropriate Technology*, 8(2), 2022. paper

## RESEARCH EXPERIENCE

**Researcher (Post-M.S.)**, Vision and Learning Lab, Kyung Hee University — Sep. 2025 – Present
Leading research on video-LLM evaluation/diagnosis and explainable video understanding; international collaborations with UC San Diego (Prof. Tsui-Wei Weng, TRACE) and Princeton University (Jihoon Chung, DeltaDirect).

**Graduate Research Assistant**, Vision and Learning Lab, Kyung Hee University — Sep. 2023 – Aug. 2025
Researched video class-incremental learning and explainable action recognition; collaborated with Intel Labs (Kyle Min) on ESSENTIAL; worked on ETRI-funded research projects.

**Undergraduate Research Intern**, Vision and Learning Lab, Kyung Hee University — Feb. 2022 – Jul. 2023
Conducted research on video action recognition; co-led CAST (NeurIPS 2023) as a co-first author.

## RESEARCH PROJECTS

**Skeleton-Based Choreography Recognition for K-POP Dance Videos** · *industry, MetaRocket* — 2026
Skeleton-based recognition of fast, multi-person choreography; KOLAS-accredited third-party performance test in progress.

**Motivational Context Analysis of Interaction-Based Reactions in Video** · *ETRI* — 2025

**Classification and Explanation Models for ASD Screening Based on Drawing Video Understanding** · *ETRI* (related publication: PCEvE) — 2024

**Deep Learning Models and Explainable AI for ASD Screening Drawing Data** · *ETRI* — 2023

**Rule-Based Group Activity Recognition Video Dataset (AI Hub)** · *national project, GAZZILABS* — 2022
Developed and validated AI recognition algorithms.

## HONORS & SCHOLARSHIPS

**Top Reviewer**, NeurIPS 2026 — 2026

**Spotlight / Highlight papers**: NeurIPS 2025 Spotlight (top 3.5%) · ICCV 2025 Highlight (top 2.9%) · CVPR 2025 Workshop XAI4CV Spotlight (top 16.7%) — 2025

**Kyung Hee Scholarship for Outstanding Undergraduate Alumni** — Sep. 2023 – Aug. 2025
Full M.S. tuition for 4 semesters (approx. $21,600 total)

**Woowa Brothers Scholarship**, JUMP Foundation (approx. $2,900) — 2022

## ACADEMIC SERVICE

**Reviewer** NeurIPS 2026 (*Top Reviewer*) · AAAI 2025 · CVPR 2026 Workshop (XAI4CV) · ICML 2026 Workshop (CompLearn)

## SKILLS & MILITARY SERVICE

**Programming** Python, C/C++, Bash · PyTorch · Linux, LaTeX · large-scale video/multimodal training and evaluation

**Military** Republic of Korea Army, Sergeant (active duty), Jul. 2018 – Mar. 2020; honorably discharged

## REFEREES

**Jinwoo Choi**
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**Kyungho Bae**
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*Co-first author, former lab colleague*